

EXPERIMENT-1

LINUX PROGRAMMING ENVIRONMENT & BASIC COMMANDS

1. Setting up of the platform

We can do C programming online or offline according to the needs. It is required for us to keep working on a Linux Environment.

Online: Cocalc.ai

Offline:

Methods: Dual boot, Single boot, Bootable Pendrive, VM

2. Virtual Machine Manager (Hypervisor)

Type-1 Baremetal - **baremetal** means the **actual physical computer/server** that runs the virtual machines.

E.g. HyperV

Type-2 Hosted - It generally means that the virtualization software runs **on top of an existing operating system**, rather than directly on the physical hardware.

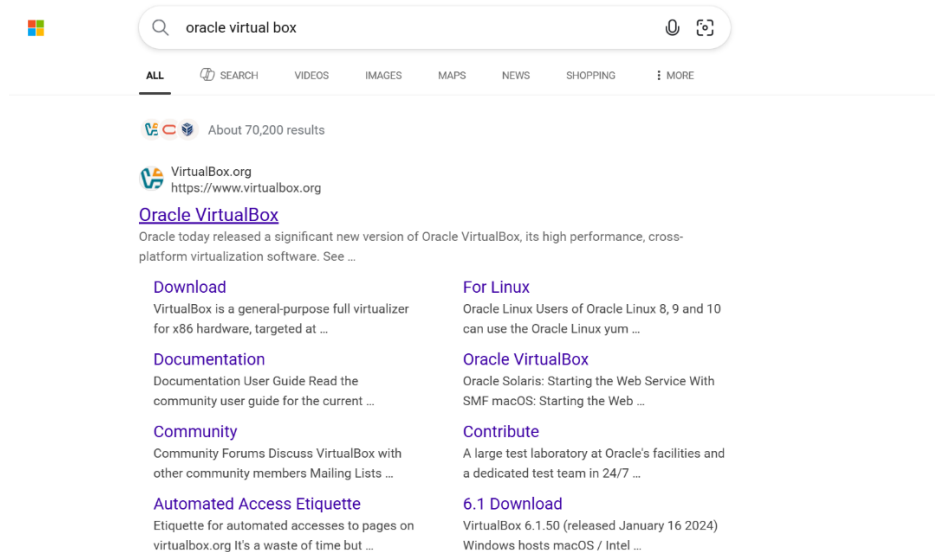
E.g Oracle Virtualbox, VM Ware.

We would be using Oracle Virtualbox.

3. Installation of Virtual Machine

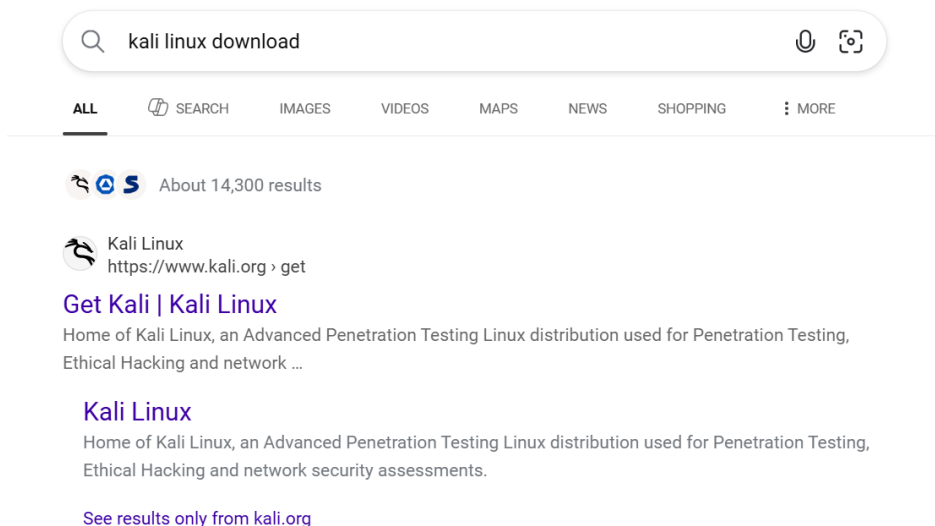
STEP-1: Download Oracle Virtual Box “windows hosts”.

Link – <https://www.virtualbox.org/>



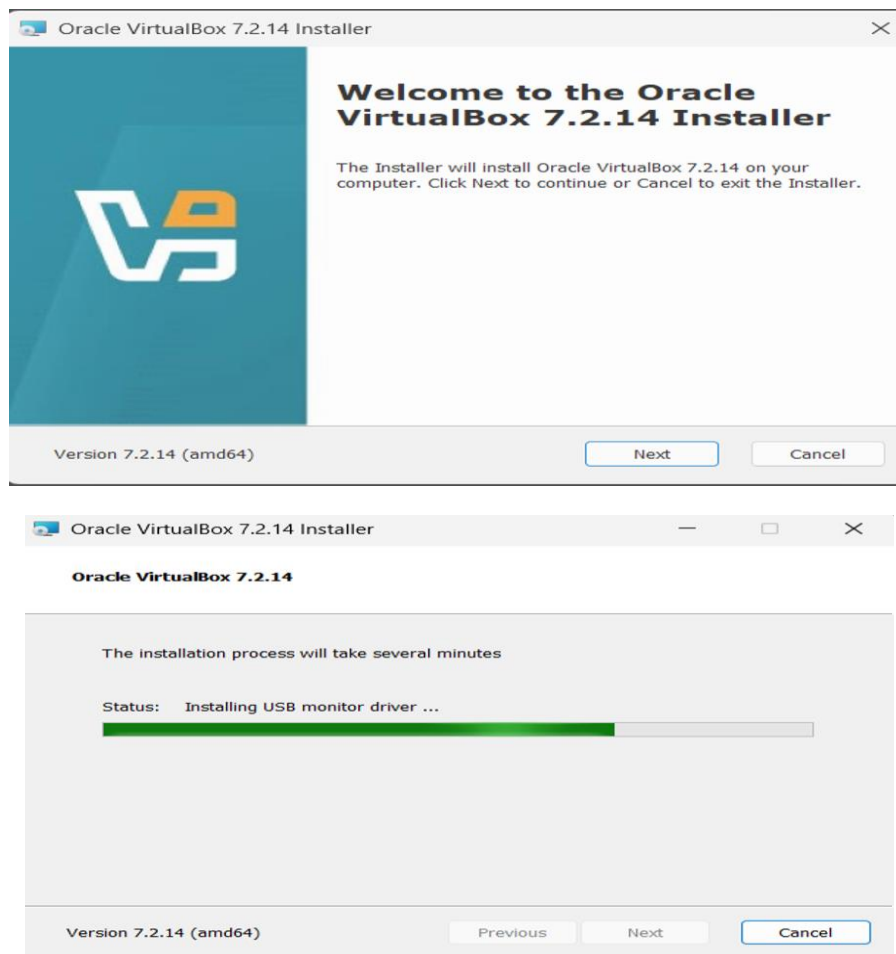
STEP- 2: Download Pre build Kali Linux Virtual Machine for Virtual box

Link - <https://www.kali.org/>

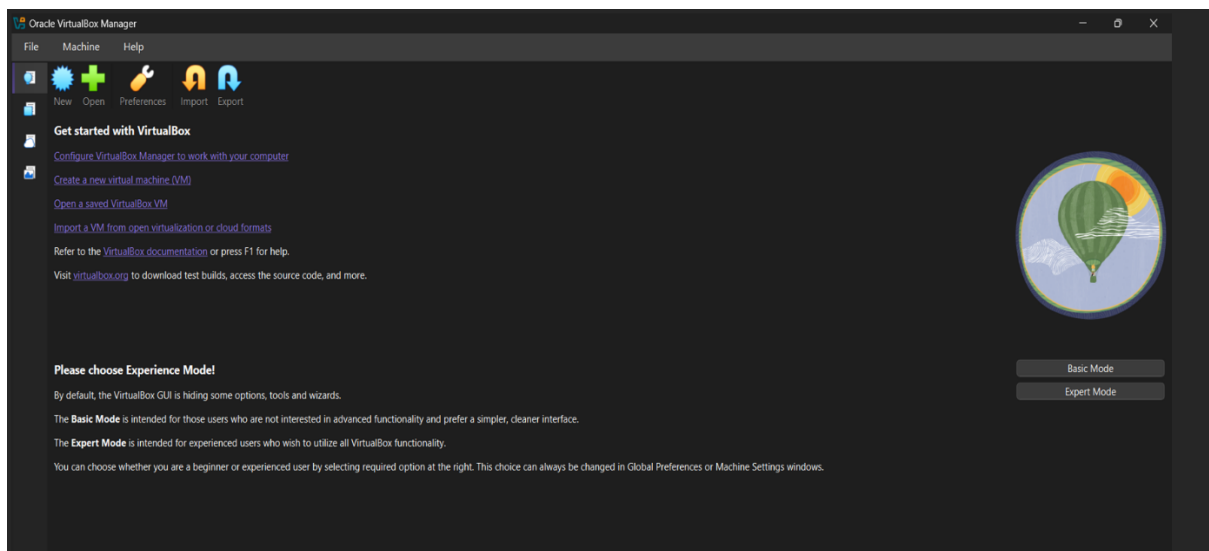


STEP-3: Extract the downloaded Kali Linux file

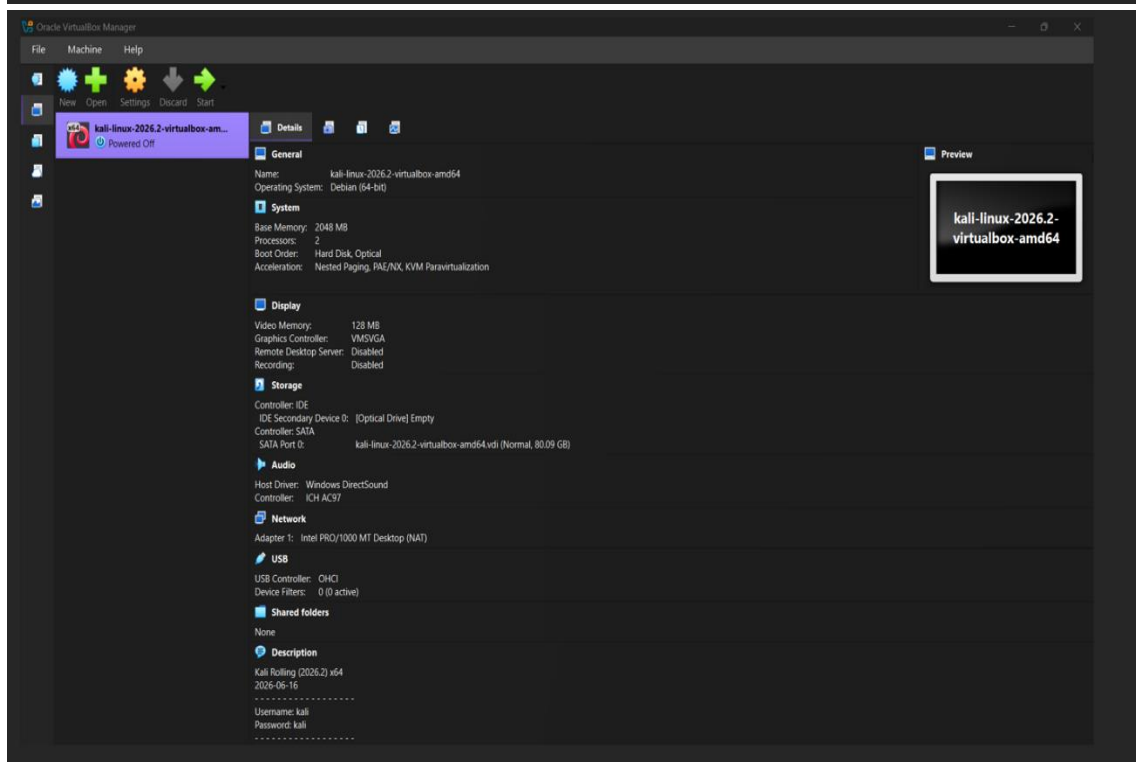
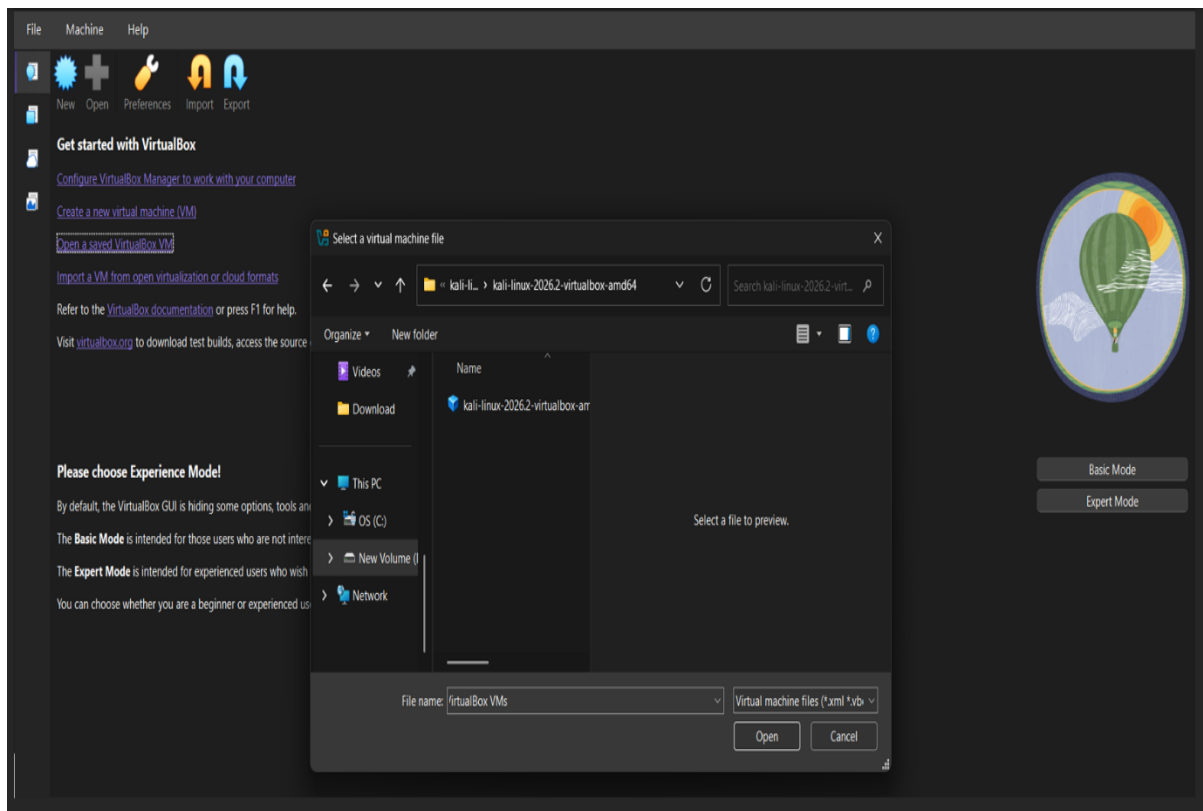
STEP-4: Install the Virtual Box



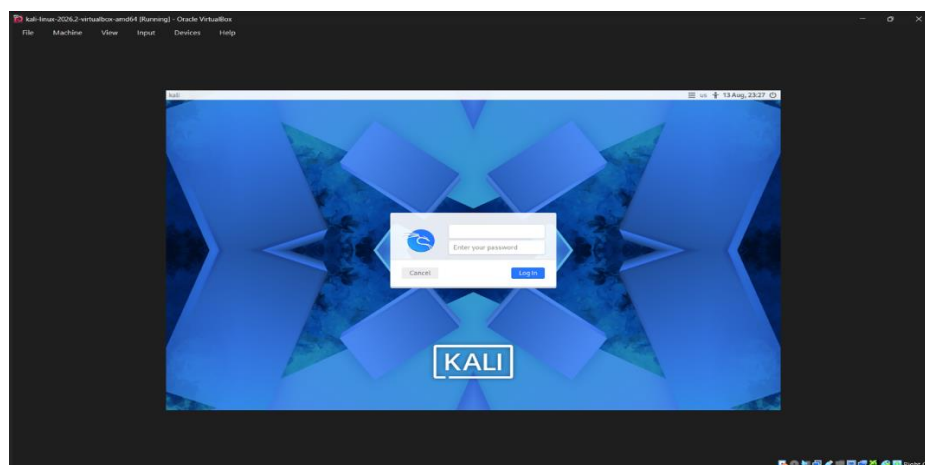
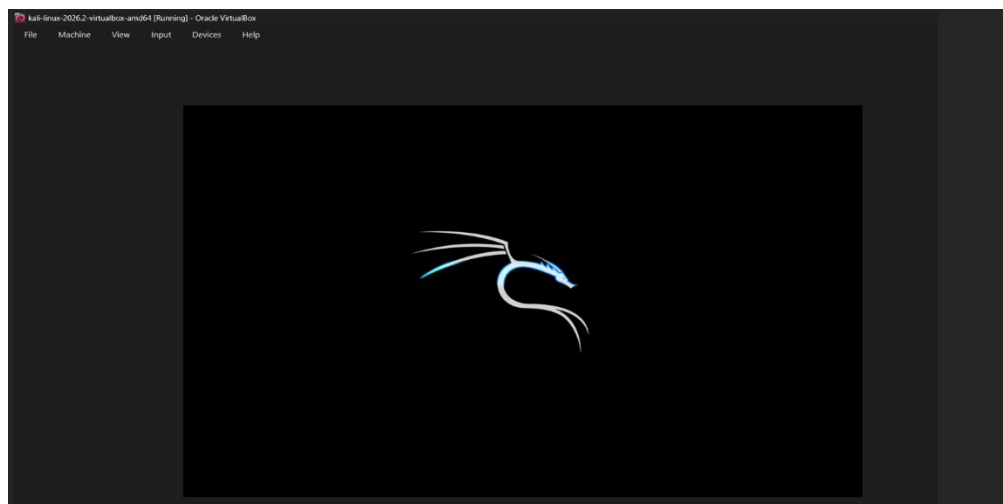
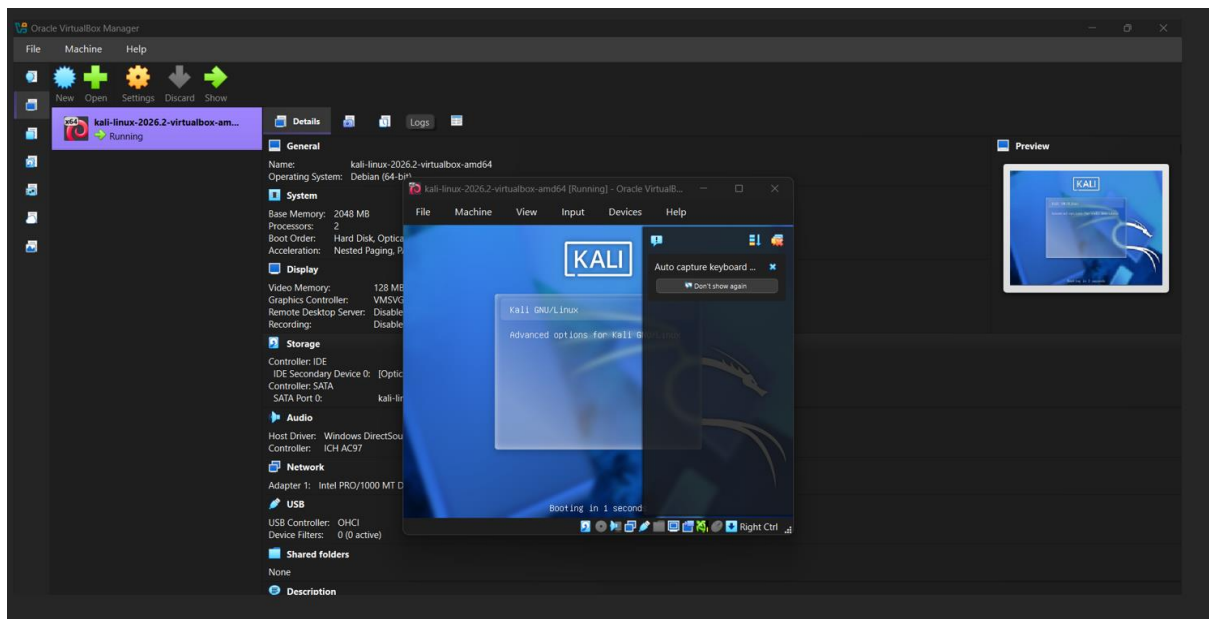
STEP-5: Open Virtual Box



STEP-6: Opening the saved Virtual Machine



STEP-7: Start the Virtual Machine



Thus ,the installation of virtual machine manager(Hypervisor Type-2 Oracle Virtual Box) & Virtual Machine Kali Linux is completed.

STEP – 8 Open terminal and create the following simple program, compile and execute to understand the working of the environment.

Working with Linux Commands :

- Terminal basics:

COMMANDS	MEANING	EXAMPLE
mkdir	Create a new directory /folder	mkdir notes.text
dir	List directories	mkdir notes.text
ls	List directories	ls
rmdir	Remove an empty directory	rmdir test
cd	Change directory	cd documents
cd..	Go up one directory	cd..
clear	Clear the terminal screen	clear
pwd	Print your current directory	pwd
History	Show all the previous used commands	history
man	Show the manual for a command	man ls
whoami	Show the current logged in username	whoami

- FILES COMMANDS:

COMMANDS	MEANING	EXAMPLE
vi	Create/edit a file	vi notes.text
file	Identify the type of a file	file notes.txt

Rm	Remove/delete a file	rm notes.txt
cp	Copy a file	cp a.txt b.txt
mv	Move or rename a file	mv a.txt b.txt
cat	Display file contents	cat notes.txt
more	View a file page by page	more notes.txt
less	View a file with the scrolling and search	less notes.txt
head	Show beginning of a file	head notes.txt
tail	Show end of a file	tail notes.txt
chmod	Change file permission	chmod 755 script.sh

- REDIRECTION COMMANDS:

>	Output redirection — sends output to a file and overwrites existing content	echo "Hello" > file.txt
>>	Append redirection — sends output to a file and adds it to the end	echo "Kali" >> file.txt
<	Input redirection — takes input from a file and gives it to a command	sort < file.txt

All the above Commands working in kali linux:

```

(kali@kali)-[~]
└─$ pwd
/home/kali

(kali@kali)-[~]
└─$ mkdir kali

(kali@kali)-[~]
└─$ rmdir thru

(kali@kali)-[~]
└─$ dir
area Desktop Documents Downloads kali Music muskan.c notes.c Pictures Projects Public sunidhi Templates tripathi Videos

(kali@kali)-[~]
└─$ ls
area Desktop Documents Downloads kali Music muskan.c notes.c Pictures Projects Public sunidhi Templates tripathi Videos

(kali@kali)-[~]
└─$ cd

```

```
Session Actions Edit View Help
(kali㉿kali)-[~]
$ whoami
kali

(kali㉿kali)-[~]
$ man
What manual page do you want?
For example, try 'man man'.

(kali㉿kali)-[~]
$ history
1  pwd
2  ls
3  makedirs thru
4  mkdir thru
5  gedit area.c
6  pwd
7  ls
8  makedirs thru
9  mkdir thru
10 gedit area.c
11 pwd
12 mkdir sunidhi
13 ls
14 #include<stdio.h>
15 vi cnotes.c
16 ped
17 pwd
18 mkdir
19 mkdir sunidhi
20 vi nidhi.c
21 pwd
```

```
(kali㉿kali)-[~/folder/folder]
$ touch file1.txt

(kali㉿kali)-[~/folder/folder]
$ cp file1.txt copy.txt

(kali㉿kali)-[~/folder/folder]
$ mv file1.txt new.txt
```

```
Session Actions Edit View Help
(kali㉿kali)-[~]
$ mkdir folder

(kali㉿kali)-[~]
$ cd folder

(kali㉿kali)-[~/folder]
$ vi file1.txt
```



```
(kali㉿kali)-[~/folder/folder]  
$ echo "hello">new.txt
```

```
(kali㉿kali)-[~/folder/folder]  
$ cat new.txt  
hello
```

```
(kali㉿kali)-[~/folder/folder]  
$ echo "welcome">>new.txt
```